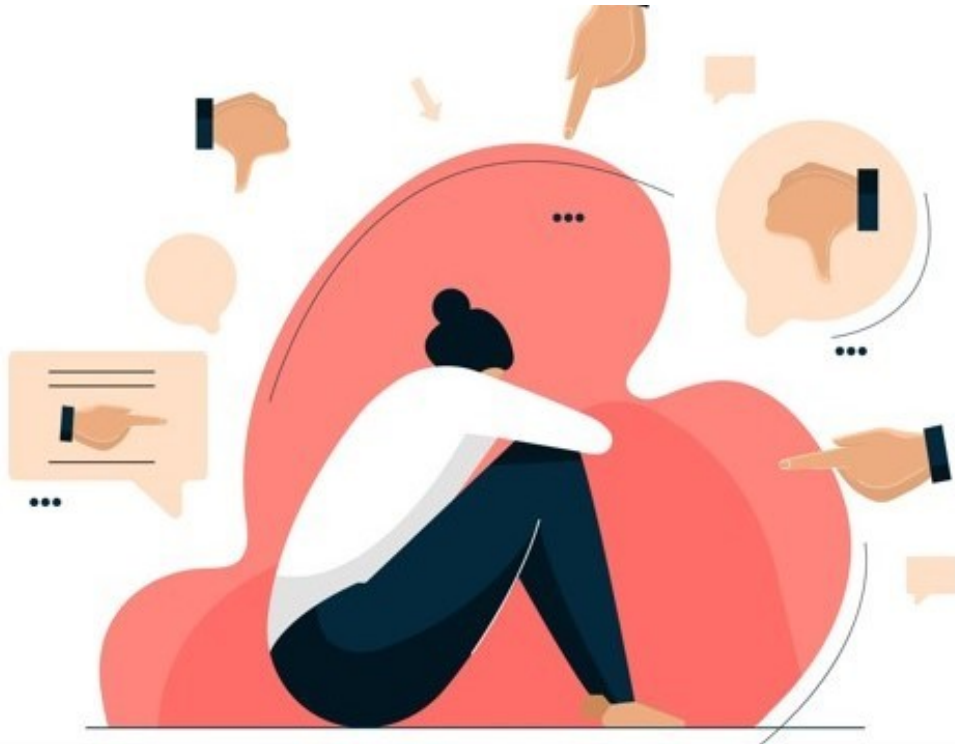


Social Media and Well being Data Analysis

Nikoleta Xenaki csd4968

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hy-484



Introduction

Σε αυτό το project θα αναλύσουμε το πως τα social media και τα activities σε αυτά επηρεάζουν το well being μας, δηλαδή τα συναισθήματα μας, ένα γεγονός που όλοι ξέρουμε πως ισχύει και εμπειρικά και λόγω διάφορων μελετών. Για αυτόν τον σκοπό χρησιμοποίησα μια βάση πραγματικών παγκόσμιων δεδομένων απο την Kaggle που συντάχθηκε το 2024 από έναν AI ερευνητή ονόματι EMİRHAN BULUT. Η ανάλυση έγινε με χρήση της Python και της βιβλιοθήκης της networkX.

Our Data Set (103 rows)

User_ID: Unique identifier for each user.

Age: Age of the user in years.

Gender: Gender identity of the user (Female, Male, Non-binary).

Platform: The social media platform used by the user (e.g., Instagram, Twitter, Facebook, LinkedIn, Snapchat, Whatsapp, Telegram).

Daily_Usage_Time (minutes): Total minutes spent daily on the specified platform.

Posts_Per_Day: Number of posts made per day.

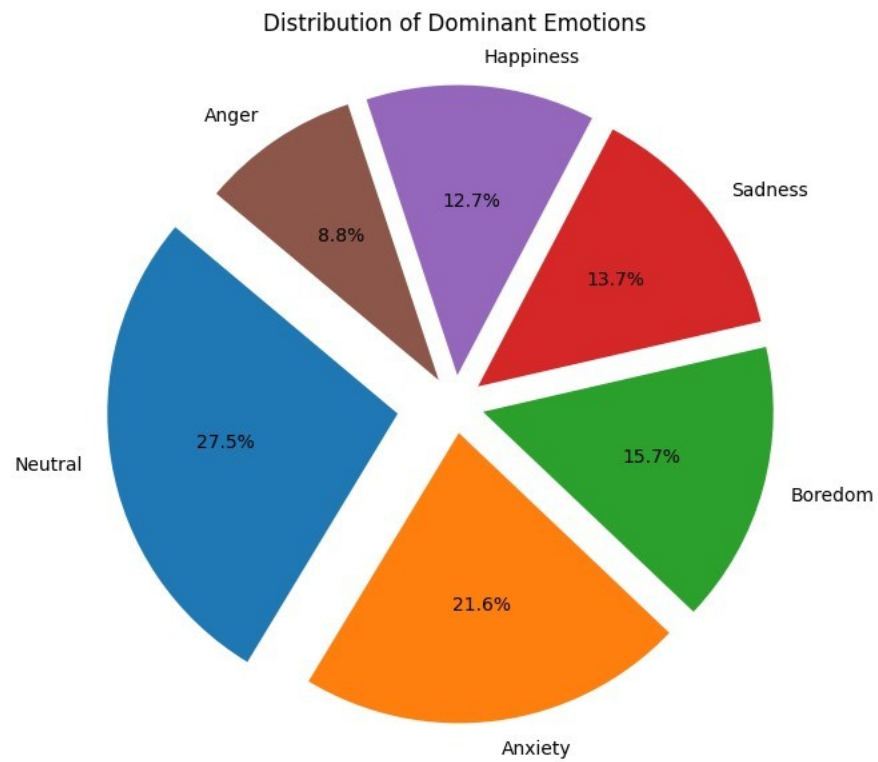
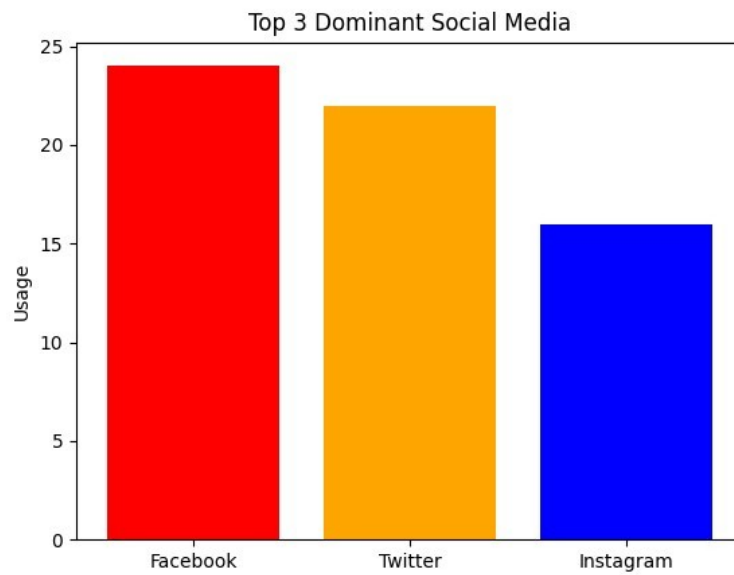
Likes_Received_Per_Day: Number of likes received per day.

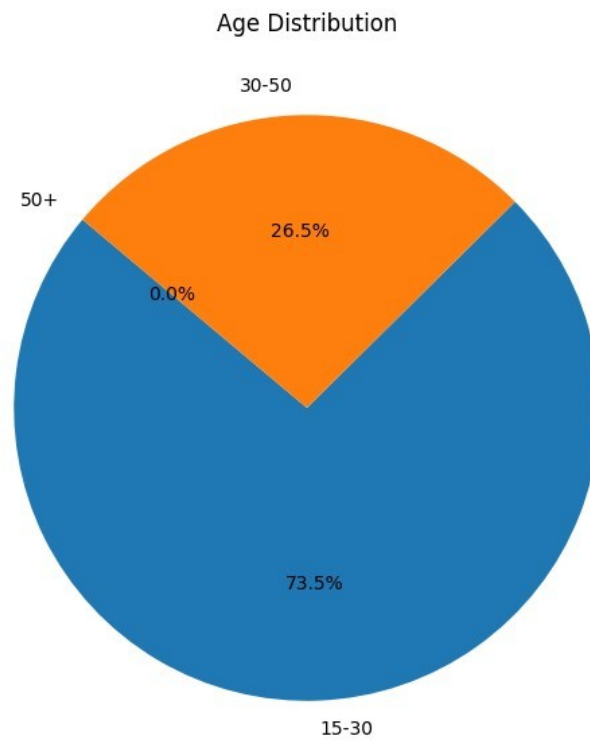
Comments_Received_Per_Day: Number of comments received per day.

Messages_Exchanged_Per_Day: Number of exchanged sent per day.

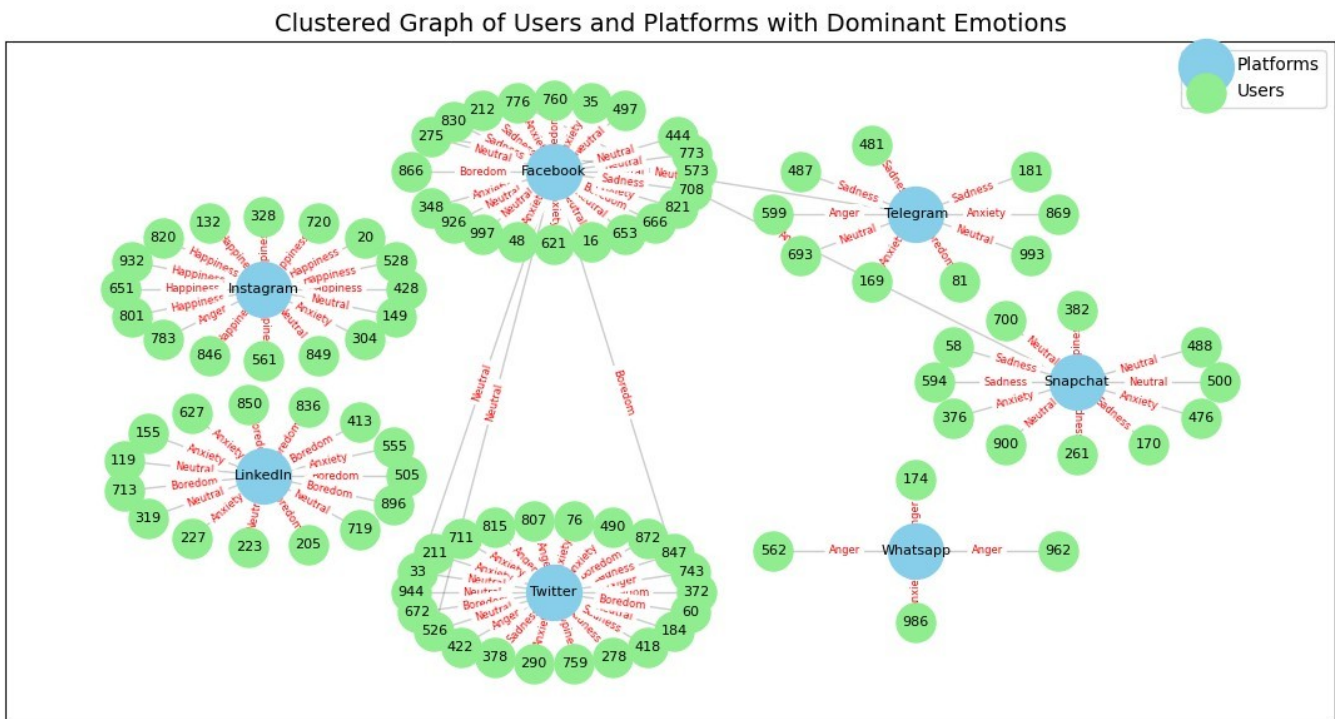
Dominant_Emotion: The dominant emotional state of the user during the day (e.g., Happiness, Sadness, Anger, Anxiety, Boredom, Neutral).

Some plots to understand our Data base





Now, let's see the social media as clusters and how users are connected to them through their emotions, to observe the overall emotions these social media provoke.



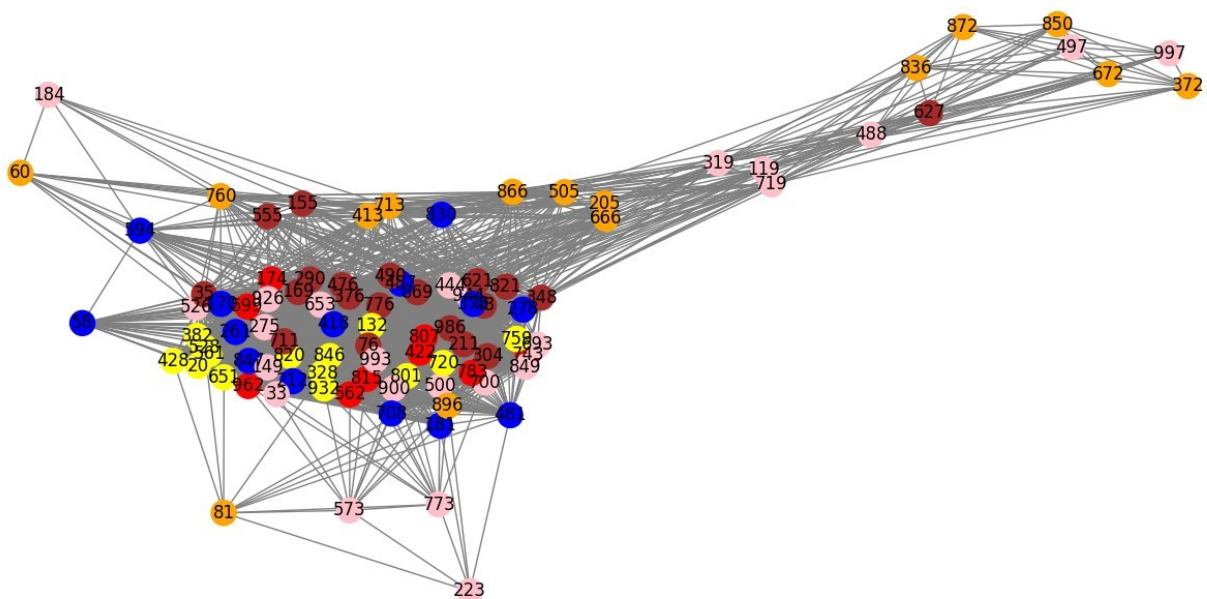
Here, we can already observe the dominant emotion/emotions for each social media. For instance, as far as Instagram is concerned, Happiness prevails, while in Facebook, Neutral feelings and feelings of anxiety hold the majority.

Emotional Homophily in terms of same activity (high/low) among users

Now, based on a hypothesis, which is also a very common phenomenon in real life, we will examine the emotional Homophily of users. We assume that users with identical/almost identical activity patterns and behaviour (high/low activity) are prone to connect with each other in terms of acquaintances or/and frianship, even if their interactions are mostly online. For this comparison the method of **Cosine Similarity** was used upon their activities and a **threshold equal to 0.9** to determine whether the result of **Cosine Similarity** indicates our preferred level of activity similarity. A real life example of such a hypothesis is when famous or/and really active social media users like influencers and youtubers tend to meet with each other somehow, online or not, to colab or just to hang out because of their same interests and activity behaviour.

Given the above, the graph below concerns individuals who are connected to each other due to their same activity patterns. The homophily index, measures how similar individuals are to each other, based on their emotions. In particular if they share positive emotions or negative ones.

Orange → Boredom , Pink → Neutral , Yellow → Happiness ,
Anger → Red , Sadness → Blue , Anxiety → Brown



Homophily index

Using this formula: $\frac{(esame - asame)}{(1 - asame)}$

(where **esame** is the number of edges in our graph where their endpoints share positive emotions (Happiness / Neutral / Boredom) and **asame** is again the number of edges where their endpoints share positive emotions but in a random distributed graph ,used as a comparison measure to our real graph.)

We got: Homophily index = 0.39

This indicates a moderate level of positive homophily, meaning that users with the same dominant emotion are more likely to connect with each other but not in extreme levels. In other words, there is indeed a tendency for people with similar emotions to interact, but it's not an overwhelmingly strong effect.

Clustering Coefficient of nodes in the previous activity graph

For computing the average clustering coefficient of a graph's node, using NetworkX, Python calculates the clustering coefficient of each node (if he has more than 2 neighbors ($k_u > 2$) , otherwise $C(u)=0$) , using this formula:

$$C(u) = \frac{2 * \text{number of triangles involving } u}{k_u * (k_u - 1)}$$

where k_u is the degree of the node u .

and afterwards:

$$\text{Average Clustering Coefficient} = \frac{\sum_{u \in V} C(u)}{V}$$

Where V is the set of nodes in the graph.

In our case

The average clustering coefficient of the previous activity graph is 0.873421107727383, meaning that users not only do they connect with others of similar activity patterns but they also form tightly knit communities due to their activities. Thus we can deduce the importance of social media in matters of human connections and interactions through social media activities.

Shortest path to Happiness and Sadness ***through social media activities per day and for any platform***

For computing the shortest path between two nodes using NetworkX, Python uses Dijkstra's algorithm since the graph is weighted with users' activities, undirected and with no negative cycles. In particular below we have two shortest paths , one for Happiness and one for Sadness, following the numbers of everyday activities and regardless of platform :

For Happiness:

Likes_Received: 33 → Messages_Exchanged: 18 → Comments_Received:20
→ Posts/Stories_Made: 3

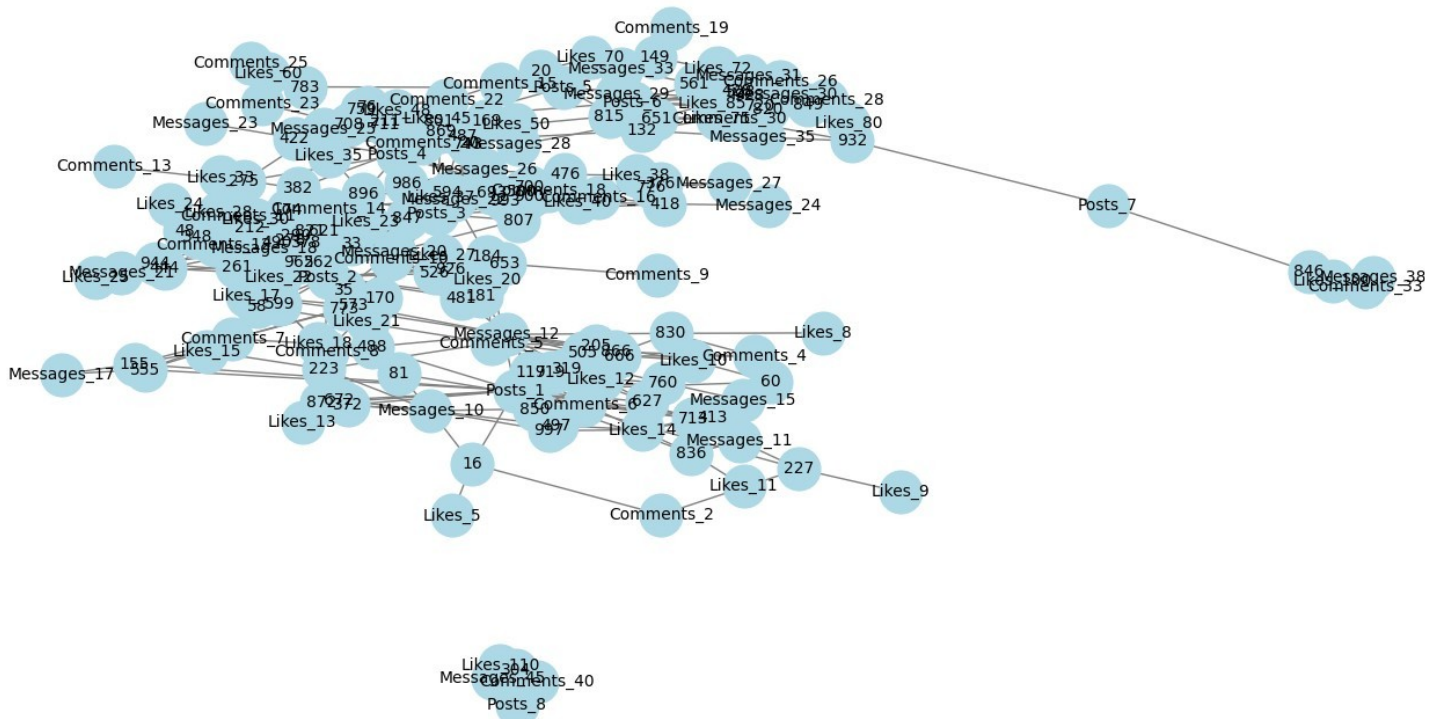
For Sadness:

Likes_Received: 8 → Messages_Exchanged: 12 → Comments_Received:4
→ Posts/Stories_Made: 1

Betweenness Centrality of nodes

In an undirected graph where users are connected with their activities, many have almost, if not exactly, the same social media activities. Nodes with the highest betweenness centralities are activity and user nodes, indicating the most important social media activities and users of the network. Apart from that, based on these important nodes, we will examine the dominant emotions these nodes hold, to see if they are mainly good/neutral emotions (happiness, neutral, boredom) or bad (anger, sadness, anxiety).

Out of 189 nodes of the graph, we will examine the nodes whose betweenness centrality is greater than 75% of all nodes (47 nodes in total). Thus we examine the most central and important nodes in the network, the ones that control the flow of information across the network.



The algorithm

For computing Betweenness Centrality of nodes, Python's NetworkX uses a variation of the **Brandes algorithm** with time complexity $O(nm)$ where n is the number of nodes and m is the number of edges. Especially for unweighted graphs the algorithm uses a shortest path counting algorithm based on BFS. Firstly for each source node are computed the shortest paths to all other nodes. Secondly dependencies for all pairs of nodes are accumulated and the betweenness contribution of each node along the shortest paths is updated iteratively. Lastly, the centrality values are normalized by the factor $(n-1)(n-2)/2$ for undirected graphs, ensuring that the values are in the range $[0, 1]$.

Results

Top 10 nodes of highest betweenness:

Posts/Stories_2, Posts/Stories_3, Posts/Stories_1, Comments_5
Comments_7, Comments_20, Posts/Stories_4, Messages_28,
Posts/Stories_6, Comments_11, Posts/Stories_5.

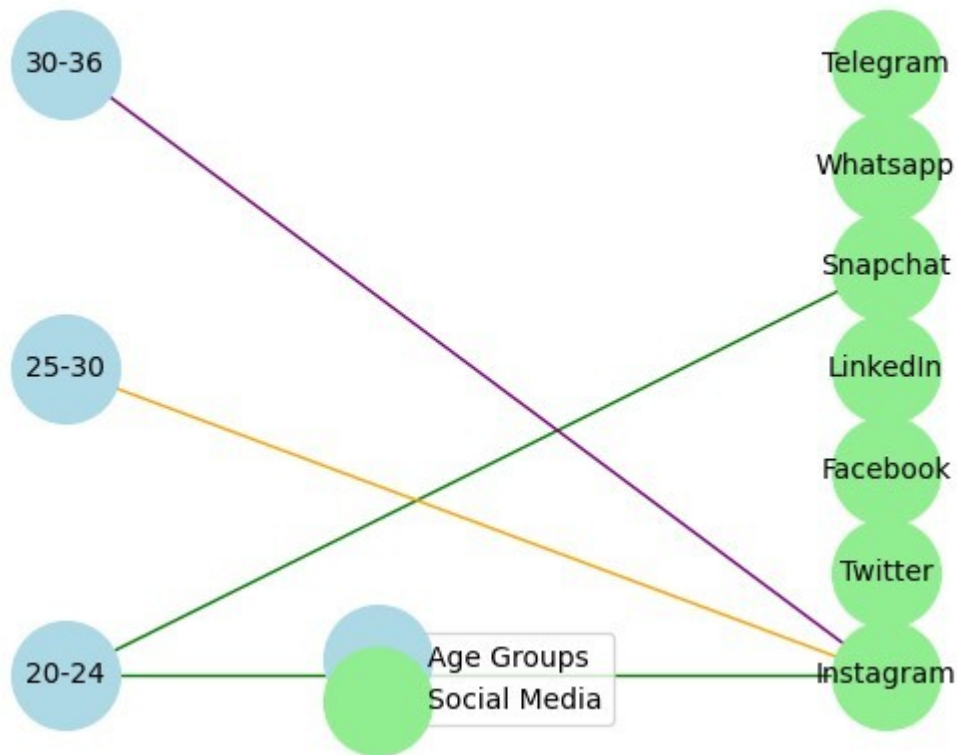
Starting with 470 points (we assume that each of these 75% important nodes hold a positive emotion so $10 \times 47 = 470$) we subtract 10 each time a node holds a negative emotion. In the end the remaining score is $250 > 470/2$ meaning that the majority of these important nodes holds positive emotions. In other words the cornerstones of this network, which control the activity flow among users, are associated with a good feedback. Such a result indicates that, in general, a person's well being is enhanced rather than deteriorated.

Age matching to Social Media according to Emotions

Now according to data set's users' ages, we match each age fluctuation with the social media, which provokes a certain emotion to the majority of these users. In this way we can see which social media platform most users turn to (and at what ages) in order to feel certain emotions.

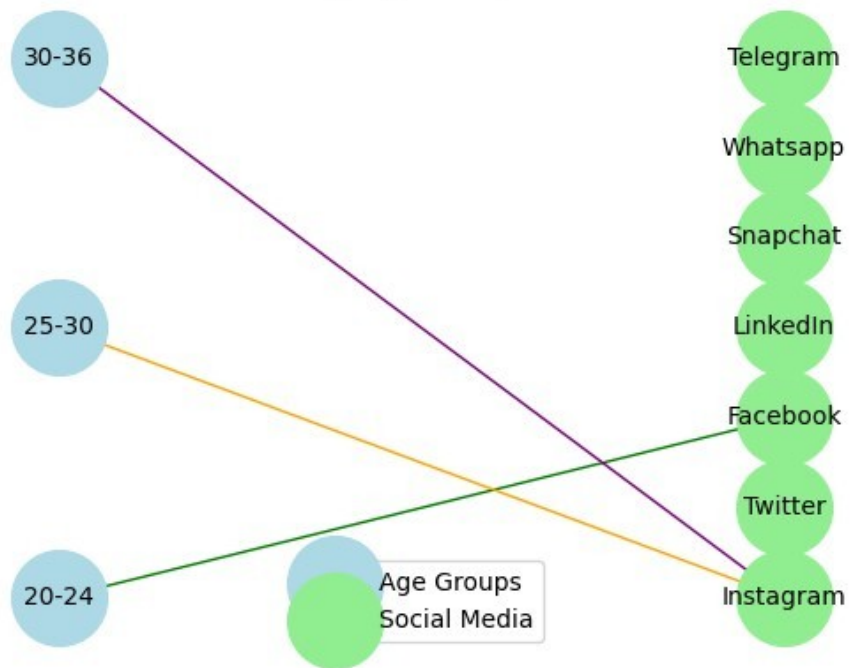
For Happiness

Social Media Preferences by Age Groups to feel Positive Emotions



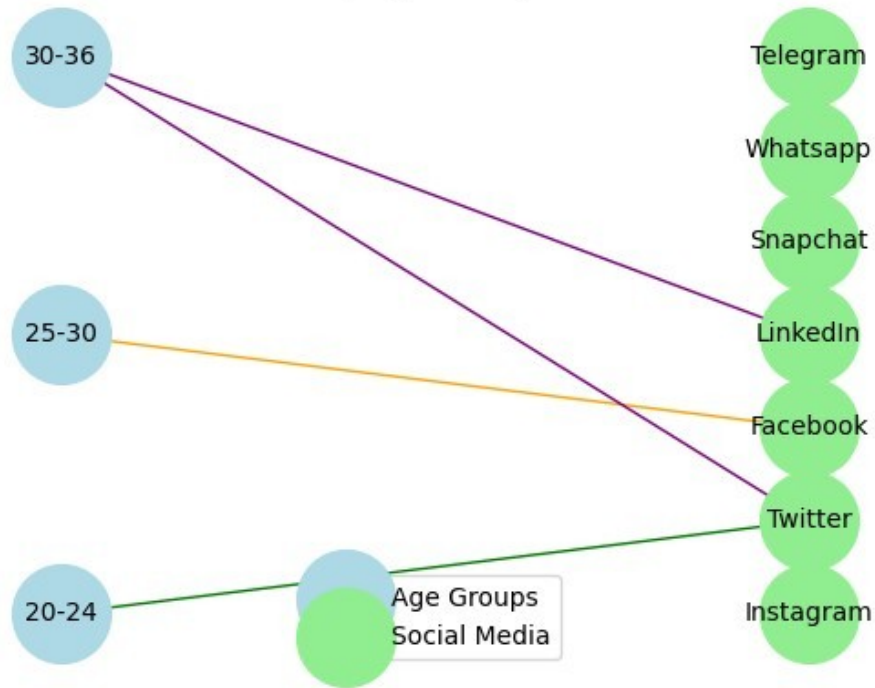
For the other two similar Positive Emotions
(Neutral,Boredom)

Social Media Preferences by Age Groups to feel Positive Emotions

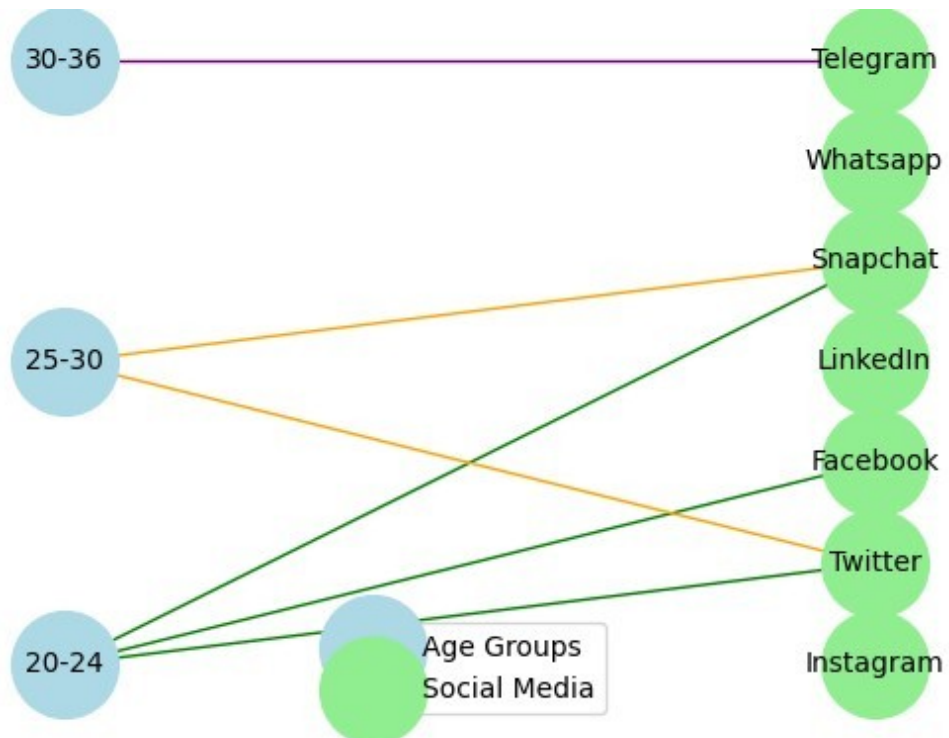


For Anxiety

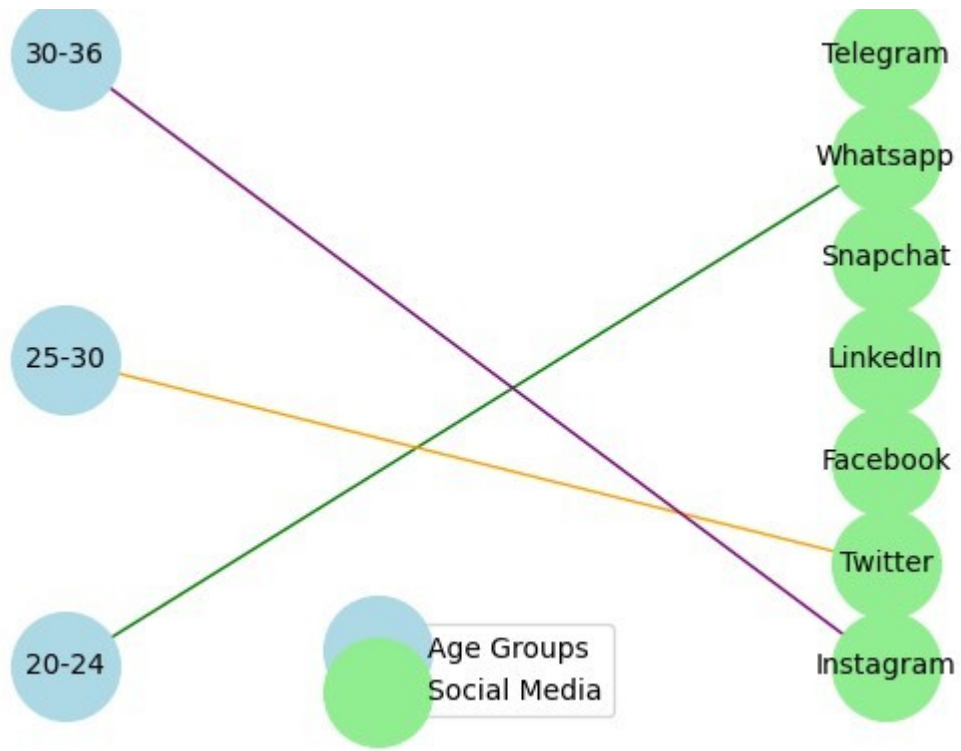
Social Media Preferences by Age Groups to feel Positive Emotions



For Sadness



For Anger



Sources



<https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book.pdf>



<https://www.kaggle.com/datasets/emirhanai/social-media-usage-and-emotional-well-being?resource=download>



Slides from Course hy-380

Σας ευχαριστώ για τον χρόνο σας!